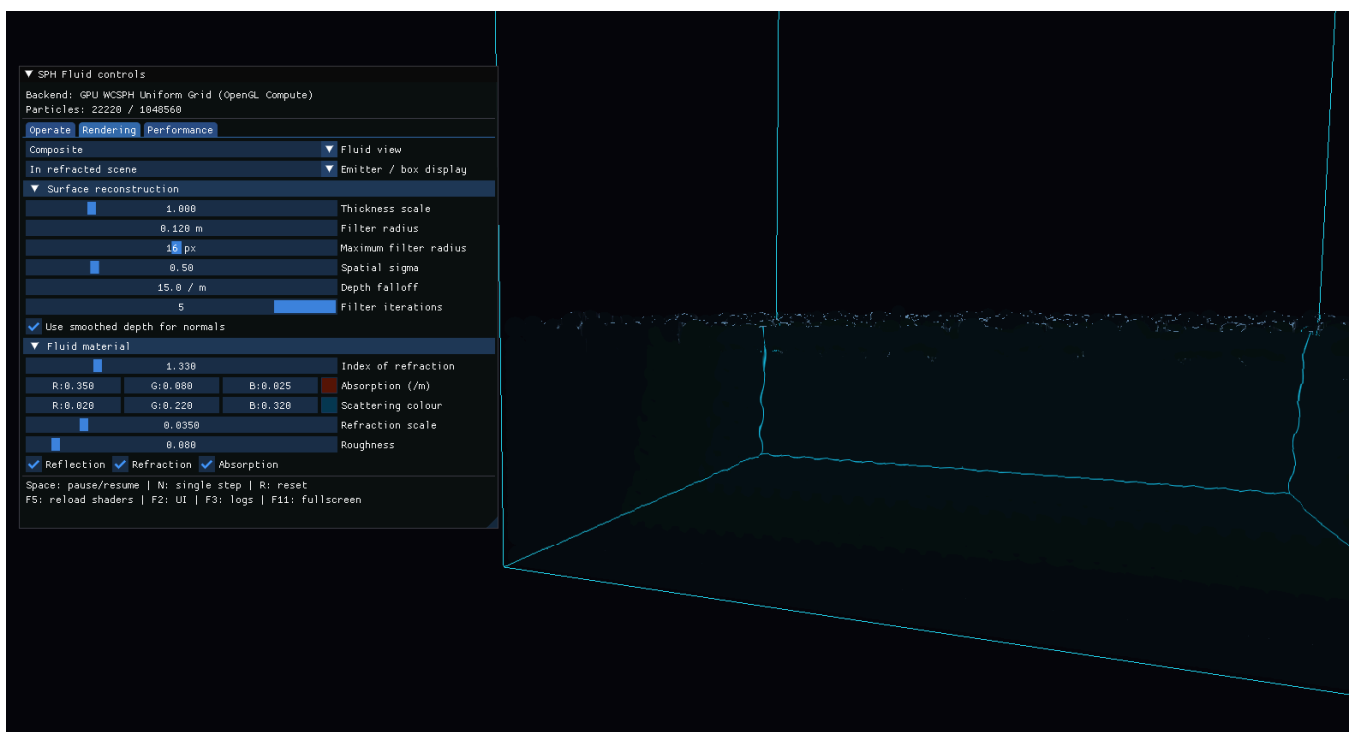
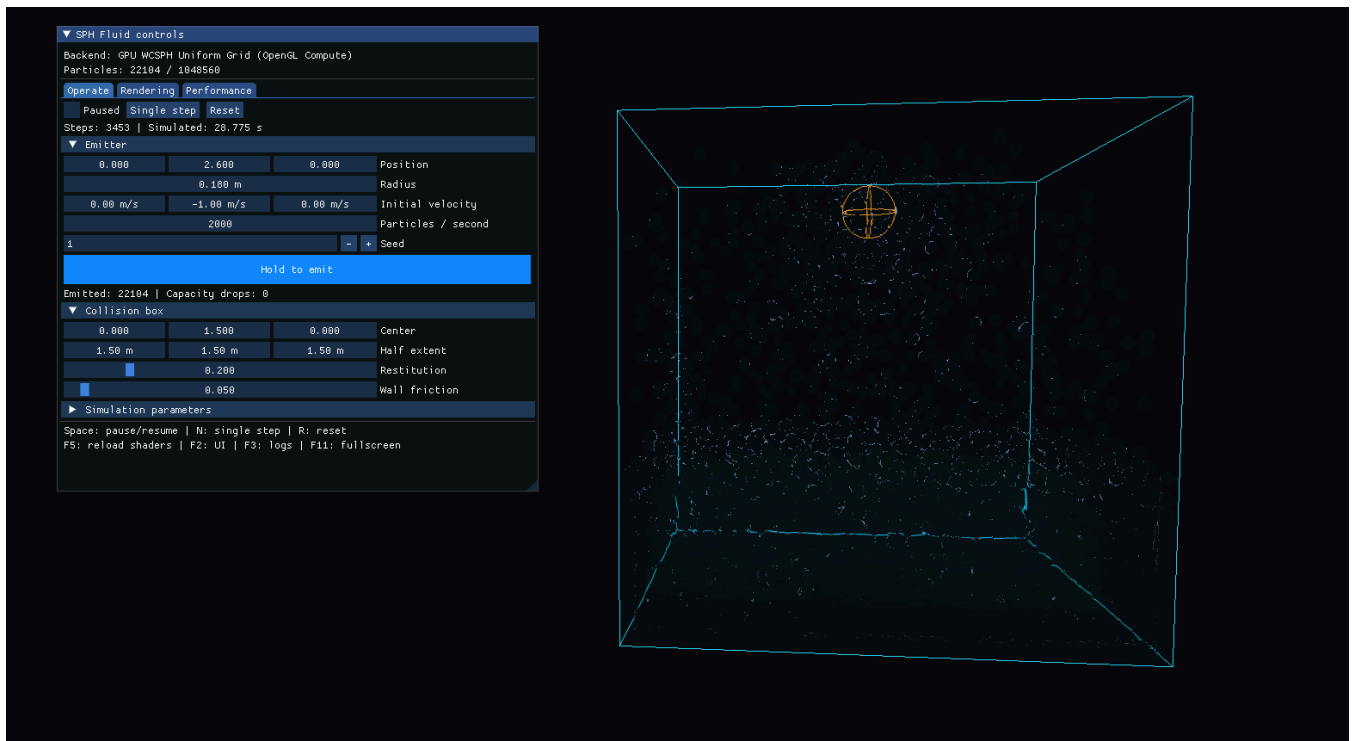




This project implements an interactive WCSPP fluid simulation in C++17 and OpenGL. It includes CPU and GPU solvers, uniform-grid neighbour search, multithreaded CPU simulation, and screen-space fluid rendering.

- [Project source](#)

- [Simulation code](#)
- [Rendering code](#)
- [Compute shaders](#)
- [Tests and benchmark results](#)

The project uses CG_Labs for window creation, input, camera control, ImGui, shader management, and CMake dependencies.

This project is built as part of CG_Labs. Follow the repository setup and Visual Studio instructions in:

- [CG Labs README](#)
- [CG Labs build guide](#)

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1. Open the CG_Labs repository folder in Visual Studio and wait for
 2. In the target dropdown next to the green Run button, select `src\SPHFluid\SPH_Fluid.exe`. Do not select an `(Install)` target.
 3. Press `F5` to build and run the project.

The ImGui interface can be used to control the particle emitter, collision box, SPH parameters, rendering mode, and performance display.

<code>Space</code>	Pause or resume the simulation
<code>N</code>	Advance one simulation step
<code>R</code>	Reset the simulation
<code>F2</code>	Show or hide the controls
<code>F5</code>	Reload shaders